

UNIT 2 – MATERIALS SCIENCE

12-Question Answer Sheet • Clean Handwritten-Style Notes

Name: _____ Roll No.: _____

Subject: Engineering Materials / Materials Science

1. Gibbs Phase Rule

The Gibbs Phase Rule gives the relationship between the number of phases, components, and degrees of freedom in an equilibrium system.

$$F = C - P + 2$$

Where F = degree of freedom, C = number of components, and P = number of phases.

For condensed systems at constant pressure: $F = C - P + 1$.

Phase: A physically homogeneous and distinct portion of a system having uniform chemical and physical properties. Example: in ice + water, ice and liquid water are two phases.

Component: The minimum number of chemically independent constituents needed to describe all phases.

Example: water is a one-component H_2O system even when ice and water coexist.

Degree of freedom: The number of independent variables such as temperature, pressure, or composition that can be changed without changing the number of phases.

2. Unary and Binary Phase Diagrams

A unary phase diagram contains one component. Its composition is fixed and it mainly shows phase changes with temperature and pressure. Example: the H_2O system.

A binary phase diagram contains two components, so composition can vary. It usually shows phase changes with temperature and composition at constant pressure. Example: the Cu-Ni system.

3. Invariant Reactions in Binary Phase Diagrams

An invariant reaction occurs at a particular temperature and composition where three phases coexist in equilibrium. For a binary condensed system, the degree of freedom is zero.

$$F = C - P + 1 = 2 - 3 + 1 = 0$$

Eutectic: one liquid transforms into two solid phases: $L \rightarrow \alpha + \beta$.

Eutectoid: one solid phase transforms into two different solid phases: $\gamma \rightarrow \alpha + \beta$. In steel: austenite $\rightarrow$ ferrite + cementite, producing pearlite.

Peritectic: a liquid and one solid phase react to form another solid phase: $L + \alpha \rightarrow \beta$.

4. Iron-Iron Carbide ($Fe-Fe_3C$) Phase Diagram

The $Fe-Fe_3C$ phase diagram describes phases and transformations in iron-carbon alloys.

Major phases: ferrite (α , BCC), austenite (γ , FCC), cementite (Fe_3C), δ -ferrite (BCC), and liquid.

Important values: eutectoid temperature $\approx 727^\circ C$ at ≈ 0.76 wt% C; eutectic temperature $\approx 1147^\circ C$ at ≈ 4.3 wt% C; maximum carbon solubility in α -ferrite ≈ 0.022 wt% C; maximum carbon solubility in γ -austenite ≈ 2.14 wt% C; cementite = 6.67 wt% C.

Sketch guide: draw temperature on the vertical axis and wt% carbon (0–6.67%) on the horizontal axis. Mark $727^\circ C$, $1147^\circ C$, 0.76% C, 4.3% C, and the α , γ , Fe_3C , liquid, and two-phase regions.

5. Cooling Curve and Transformation of Eutectoid Steel

Eutectoid steel contains approximately 0.76% carbon. When slowly cooled from the austenite region, it remains as austenite (γ , FCC) until the eutectoid temperature, about $727^\circ C$.

At $727^\circ C$, the eutectoid reaction occurs: $\gamma \rightarrow \alpha + Fe_3C$.

Ferrite and cementite form together in alternating layers, producing pearlite. Therefore, slowly cooled eutectoid steel is essentially pearlite at room temperature.

During the transformation there is a temperature arrest on the cooling curve because heat is released during the transformation.

6. Time-Temperature-Transformation (TTT) Diagram

A TTT diagram shows how austenite transforms into different microstructures as a function of temperature and time. It is also called an isothermal transformation diagram.

Typical transformation products are pearlite, bainite, and martensite.

The nose is the part of the transformation curve closest to the vertical temperature axis. It represents a region where austenite transforms particularly rapidly.

The critical cooling rate is the minimum cooling rate needed to avoid the nose and obtain martensite during rapid cooling.

7. White Cast Iron vs Gray Cast Iron

White cast iron: carbon is mainly present as cementite (Fe_3C). It has a white, bright fracture appearance, a hard and brittle microstructure, poor machinability, and high wear resistance.

Gray cast iron: carbon is mainly present as graphite flakes. It has a gray fracture appearance, a graphite-flake microstructure in a ferrite/pearlite matrix, better machinability, and good vibration-damping capacity.

8. Pearlite, Bainite, and Martensite

Pearlite: forms during relatively slow/moderate cooling; consists of ferrite and cementite in a layered structure; it has moderate hardness and relatively good ductility.

Bainite: forms at an intermediate transformation temperature; consists of ferrite with carbides; it is harder than pearlite and has moderate ductility.

Martensite: forms by very rapid cooling (quenching); it is a supersaturated carbon structure in iron with a BCT crystal structure; it has very high hardness but low ductility.

9. Effect of Alloying Elements on Steel

Alloying elements can change critical transformation temperatures, transformation kinetics, and the hardenability of steel.

Elements such as Ni, Mn, Cr, Mo, and Si can shift transformation temperatures and alter the stability of ferrite and austenite.

Hardenability is the ability of steel to form martensite to greater depth during quenching. Many alloying elements increase hardenability by slowing the transformation of austenite into pearlite and bainite.

Examples: Cr increases hardenability and wear resistance; Mo increases hardenability; Mn increases hardenability; Ni increases hardenability and toughness.

10. Binary Isomorphous vs Binary Eutectic Phase Diagrams

A binary isomorphous system has complete solid solubility between the two components. Example: Cu-Ni. It has liquid, liquid + solid-solution, and solid-solution regions.

A binary eutectic system has limited/partial solid solubility and contains a eutectic point where liquid transforms into two solid phases. Example: Pb-Sn.

General eutectic reaction: $L \rightarrow \alpha + \beta$.

Sketch guide: isomorphous: one continuous solid-solution region below a liquidus/solidus pair. Eutectic: two liquidus branches meeting at the eutectic point, with $\alpha + \beta$ below.

11. Partial Solid Solubility vs Complete Solid Insolubility

In a binary eutectic system with partial solid solubility, the two solid phases can dissolve a limited amount of each other. The diagram therefore contains α and β solid-solution regions.

With complete solid insolubility, the components have essentially no mutual solubility in the solid state. The solid phases are therefore nearly pure A and pure B.

The main structural difference is that partial solubility gives α and β solid solutions, whereas complete insolubility gives essentially pure solid components.

Sketch guide: partial solubility: show α and β terminal solid-solution regions around the eutectic. Complete insolubility: show essentially pure A and B solid phases separated by the eutectic composition.

12. Classification of Cast Irons by Carbon Form

Cast irons can be classified according to the form and distribution of carbon in their microstructure:

Gray cast iron: graphite flakes.

White cast iron: cementite (Fe_3C).

Malleable cast iron: temper carbon produced by heat treatment of white cast iron.

Ductile/Nodular cast iron: graphite nodules or spheroids.

The shape and distribution of carbon influence hardness, ductility, toughness, machinability, and damping capacity.